

Aththota Gamage Pramoda Sahan Kumari

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PROFESSIONAL SUMMARY

Dynamic researcher and educator with over a decade of experience spanning biochemistry, molecular biology, and computational chemistry. Proven track record of impactful research at world-class institutions, including **Wayne State University**, **Boston Children's Hospital**, **Harvard Medical School**, and **Brigham Young University**. Extensive hands-on expertise in cancer research, immunology, and advanced molecular biology techniques, complemented by strong computational skills in molecular modeling and drug design. Adept at bridging experimental and computational approaches to accelerate translational research, with a strong publication record and commitment to driving breakthroughs in the health sector.

EDUCATION

PhD Candidate – Computational Chemistry

Wayne State University,

August 2023 to present.
Michigan, United States.

Master of Science – Biochemistry

Brigham Young University,
Awarded a full-tuition scholarship.

April 2021
Utah, United States

Bachelor of Science Honors in Biochemistry and Molecular Biology

University of Colombo,

March 2016
Sri Lanka

RESEARCH TECHNIQUES

- **Computational Methods:** Molecular Modeling, Molecular Docking (Structure-Based Drug Design), Free Energy Calculations, Molecular Dynamics Simulations, Quantum Chemistry, QM/MM Methods.
- **Cell & Molecular Biology:** Cell Culture, Immunohistochemistry, Immunocytochemistry, Western Blotting, Immunoprecipitation, CRISPR Cas9 to do targeted mutations, siRNA/shRNA Transfections.
- **Genomics & Sequencing:** Single-cell RNA Sequencing Analysis, 16S rRNA Sequencing, Real-time & Conventional PCR, Gel Electrophoresis, DNA Extraction.
- **Immunology Techniques:** FACS, Isolation of Immune Cells, Immunofluorescence Staining.
- **Other:** Protein Assays (Bradford, PI3K), Expression Cloning, Viral Transfections, SDS-PAGE, Bacterial Culturing.

RESEARCH AND WORKING EXPERIENCE

Graduate Assistant (Research and Teaching)

Wayne State University

August 2023 to present
Michigan, United States

- Conducted computational analysis of drug-mediated modulation of **Amyloid- β aggregation** in Alzheimer's disease.
- Performing **Classical**, **QM/MM** simulations and **Machine Learning** of fluorescent proteins (mStayGold, mBaojin) to study structural and functional properties.
- Developing expertise in **computational techniques**, including advanced molecular simulations, machine learning applications, and bioinformatics workflows.
- Led **Organic Chemistry discussion sessions**, facilitated weekly office hours, and evaluated student performance through quizzes and exams.

- Instructed **Biochemistry and Organic Chemistry laboratory classes**, ensuring adherence to safety protocols and fostering hands-on learning.

Researcher

May 2021 to June

2023

Boston Children's Hospital and Harvard Medical School

Boston, United States

- Investigated **NRP2 signaling pathways in immunity**, contributing to translational research in immunology.
- Applied advanced molecular biology techniques, including **Western blotting, immunoprecipitation, immunohistochemistry, immunocytochemistry, and single-cell RNA sequencing analysis**.
- Executed **siRNA/shRNA transfections** and performed mammalian cell culture and cell-based assays to support experimental objectives.
- Acquired hands-on experience in **immunology**, including isolation of immune cells (PBMCs, CD4+, CD8+, and regulatory T cells) from human and mouse samples, in vitro cell culture assays, and standard readouts such as **FACS, ELISA, and qPCR**.
- Performed **DNA and RNA extractions**, gel electrophoresis, and both real-time and conventional PCR to support molecular biology research.
- Conducted data analysis and visualization, including statistical calculations using **GraphPad Prism**.

Graduate Research Assistant- Biochemistry

September 2019 to April 2021

Fritz B. Burns Cancer Research Laboratory, Brigham Young University

Utah, United States

- Conducted independent research on **SGK2, 14-3-3, and HUWE1 regulation of PTOV1 stability in cancer**, and contributed to collaborative studies on the **oncogenic capacity of TNK1 clinical mutants**.
- Gained extensive wet lab experience in **biochemistry, molecular biology, and cell-based assays**, supporting diagnostic solutions for prostate cancer treatment.
- Utilized advanced techniques including **RNAi interference, transient transfection, enzymatic assays, CRISPR/Cas9 genomic editing, plasmid subcloning, and PCR**; performed mammalian cell culture, live-cell imaging (Incucyte), and viral vector work in ML-2 laboratory settings.
- Designed and executed experimental studies, analyzed qualitative and quantitative data, and prepared comprehensive reports, tables, and graphs summarizing research findings.
- Delivered presentations at scientific conferences, demonstrating strong communication and data visualization skills.

Graduate Teaching Assistant

September 2019 to April 2021

Department of Chemistry & Biochemistry, Brigham Young University.

Utah, United States

- Delivered **Biochemistry and Organic Chemistry lectures, laboratory sessions, and tutorials** for first- and second-year undergraduate students, employing effective instructional techniques to enhance engagement and learning outcomes.
- Conducted **special assessments and personalized tutoring sessions** to support individual student needs and academic success

Research Assistant- Molecular microbiology

July 2016 to September 2019

Department of Microbiology, University of Sri Jayewardenepura

Sri Lanka.

- Conducted independent research on **“Determining the role of the gut microbiome in the etiology of colorectal cancer in Sri Lanka”**, including patient sample collection, clinical data acquisition, and execution of all wet-lab experiments.

- Presented findings at **local and international conferences**, demonstrating strong communication and scientific reporting skills.
- Mentored undergraduate students on laboratory protocols and experimental techniques, fostering academic development and hands-on learning.

PUBLICATIONS – REFERENCE JOURNAL ARTICLES

1. Mark A. Hix, **A. G. Pramoda Sahankumari**, Ashok S. Bhagwat and Alice R. Walker, “DNA hairpin base-flipping dynamics drives APOBEC3A recognition and selectivity,” *Physical Chemistry Chemical Physics*, 2025, 27, 11551-11557. DOI: 10.1039/D5CP00321K -**Impact Factor 2.9**
2. Gamage, B. D., Ranasinghe, D., **Sahankumari, A. G. P.**, & Malavige, G. N. (2024). Metagenomic analysis of colonic tissue and stool microbiome in patients with colorectal cancer in a South Asian population. *BMC Cancer*, 2024, 1124. <https://doi.org/10.1186/s12885-024-12885-4> - **Impact Factor 3.46**
3. Timna Agur, Johannes Wedel, Sayantan Bose, **AG Pramoda Sahankumari**, Daniel Goodman, Sek Wong Kong, Chandra C. Ghosh and David M. Briscoe “Inhibition of Mevalonate metabolism by statins augments the immunoregulatory phenotype of vascular endothelial cells and inhibits the costimulation of CD4+ T cells” – *Am J Transplant* 2022 Mar;22(3):947-954. doi: 10.1111/ajt.16872. Epub 2021 Nov 8 – **Impact Factor F 8.08**
4. Pennington KL, McEwan C, Woods J, Muir C, **Sahankumari AGP**, Eastmond R, Egbert CM, Heaton T, Piccolo SR, Andersen JL “SGK2, 14-3-3 and HUWE1 coordinately regulate PTOV1” –*Molecular Cancer Research* – **Impact Factor 5.852** (Manuscript DOI: 10.1158/1541-7786.MCR-20-1076 Manuscript ID: MCR-20-1076R2)
5. **AGP Sahankumari**, B.D Gamage, G.N. Malavige. “Patterns of gut microbiota in a South Asian population of colorectal cancer” – bioRxiv preprint (doi: <https://doi.org/10.1101/694125>)

RESEARCH ABSTRACTS AND CONFERENCE PRESENTATIONS

1. **A.G.P Sahankumari**, Alice R Walker, Simulations of Excited-State Dynamics of mStayGold. American Chemical Society (ACS) Fall National Meeting, August 17–21, 2025. (Poster presentation)
2. **A.G.P Sahankumari**, Alice R Walker, Excited-State Simulations of mBaojin. 55th Midwest Theoretical Chemistry Conference (MWTCC55), Wayne State University, Detroit, MI, May 29–31, 2025. (Poster presentation)
3. **A.G.P Sahankumari**, Alice R Walker, Determining the Molecular Mechanism of mBaojin: Photostability and Chloride Sensitivity. 2025 Graduate Research Symposium, Wayne State University, Detroit, MI, February 26, 2025. (Poster presentation)
4. **A.G.P Sahankumari**, Alice R Walker, Simulation of Excited-State Dynamics of mStayGold. 2025 Chemistry Graduate Research Symposium, Wayne State University, Detroit, MI, October 24, 2025. (Poster presentation)
5. **A.G.P Sahankumari**, B.D Gamage, G.N. Malavige. Determining the specific gut microbiota patterns associated with colorectal cancer. The Supplement of the *Journal of Gastroenterology and Hepatology*. Asian Pacific Digestive Week 2018 (APDW 2018), Coex, Seoul, South Korea. – (E – poster presentation)
6. **A.G.P Sahankumari**, B.D Gamage, G.N. Malavige. Determining the gut microbiota patterns associated with early-stage CRC and late-stage CRC. *The Sri Lanka Journal of Surgery* August 2018 Volume 36, Issue Supplement S1 ISSN 1391-491X. (Oral presentation)
7. **A.G.P Sahankumari**, B.D Gamage, G.N. Malavige. Identification of the gut microbiota patterns

associated with Diabetes mellitus (2018). The Ceylon Medical Journal Volume 63 Supplement 1, July 2018. (Oral presentation)

8. **A.G.P S. Gamage**, B.D Gamage, G.N. Malavige. Determining the role of the gut microbiome in the etiology of colorectal cancer. In: Proceeding of the Ceylon College of Physicians, 50th Annual Academic Sessions; 2017 Sep 6-9; Colombo, Sri Lanka: CCP; 2017. Abstract nr 61 (Oral and poster presentation).
9. Katie L. Pennington, James Woods, Colin Muir, Colten M. McEwan, **Pramoda S. K. Aththota Gamage**, Riley J. Eastmond, Crissy M. Egbert, Tyler Heaton, Stephen R. Piccolo, Joshua L. Andersen. 14-3-3 binding during G1 stabilizes and sequesters PTOV1 to the cytoplasm. American Association for cancer research, Annual meeting 2021, United States.
10. Katie L. Pennington, James Woods, Colin Muir, Colten M. McEwan, **Pramoda S. K. Aththota Gamage**, Riley J. Eastmond, Crissy M. Egbert, Tyler Heaton, Stephen R. Piccolo, Joshua L. Andersen. 14-3-3 binding stabilizes and sequesters PTOV1 to the cytoplasm during G1” The journal of Biological Chemistry, 2023, Vol.299(3) p.104286, Article 104286

CERTIFICATE COURSES AND WORKSHOPS

- Biomedical Responsible Conduct of Research (CITI Program, A Division of BRANY, United States. Credential ID – 42513179)
- Human Research Group 1Biomedical (CITI Program, A Division of BRANY, United States. Credential ID – 42513178).
- Certificate Course on Quantitative/Qualitative data analysis (University of Sri Jayewardenepura, Sri Lanka.).

MEMBERSHIPS

- Member of American Association for Cancer Research, USA.
- Member of Chemical Society, University of Colombo, Sri Lanka